

OSSP - PRACTICAL ASSIGNMENT - 5

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1.

```
1 #include <unistd.h>
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <time.h>
5 #include <sys/wait.h>
6 int main()
7 {
8     int fd[2];
9     int pid, data;
10
11     double start, end, timetaken;
12     if (pipe(fd) == -1)
13     {
14         perror("Pipe failed");
15         return 1;
16     }
17     pid = fork();
18     start = clock();
19
20     if (pid == 0)
21     {
22         printf("Consumer\n");
23         close(fd[1]);
24         for (int i = 0; i < 5; i++)
25         {
26             read(fd[0], &data, sizeof(data));
27             printf("Consumed %d\n", data);
28         }
29         close(fd[0]);
30     }
31     else
32     {
33         printf("Producer\n");
34         close(fd[0]);
35         for (int i = 1; i <= 5; i++)
36         {
37             data = i * 16;
38             write(fd[1], &data, sizeof(data));
39             printf("Produced %d\n", data);
40         }
41         close(fd[1]);
42         wait(NULL);
43     }
44     end = clock();
45 }
```

```
haadi_hasil@DESKTOP-J6ARPCK:~$ gedit producerconsumer.c
haadi_hasil@DESKTOP-J6ARPCK:~$ gcc producerconsumer.c
haadi_hasil@DESKTOP-J6ARPCK:~$ ./a.out
Producer
Produced 16
Produced 32
Produced 48
Produced 64
Produced 80
Consumer
Consumed 16
Consumed 32
Consumed 48
Consumed 64
Consumed 80
Communication time is 0.000121 seconds
Communication time is 0.000125 seconds
```

2.

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <stdlib.h>
4
5 int main()
6 {
7     int fd[2];
8
9     if (pipe(fd) == -1)
10    {
11        perror("pipe");
12        return 1;
13    }
14
15    int pid = fork();
16
17    if (pid > 0)
18    {
19        close(fd[0]);
20
21        dup2(fd[1], STDOUT_FILENO);
22        close(fd[1]);
23
24        execlp("ls", "ls", "-l", NULL);
25
26        perror("execlp ls");
27    }
28    else if (pid == 0)
29    {
30        close(fd[1]);
31        dup2(fd[0], STDIN_FILENO);
32        close(fd[0]);
33
34        execlp("grep", "grep", ".c", NULL);
35
36        perror("execlp grep");
37    }
38    else
39    {
40        perror("fork");
41        return 1;
42    }
43
44    return 0;
45 }
```

```

haadi_hasil@DESKTOP-J6ARPCK:~$ gedit program5.c
haadi_hasil@DESKTOP-J6ARPCK:~$ gcc program5.c
haadi_hasil@DESKTOP-J6ARPCK:~$ ./a.out
-rw-r--r-- 1 haadi_hasil haadi_hasil 392 Sep 1 05:37 FCFS.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 503 Sep 1 05:40 SJF.c
-rw-r-xr-x 1 haadi_hasil haadi_hasil 12 Jul 30 06:00 ac
-rwxr-xr-x 1 haadi_hasil haadi_hasil 16216 Sep 15 06:59 client
-rw-r--r-- 1 haadi_hasil haadi_hasil 312 Sep 15 06:50 client.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 230 Sep 15 06:36 client1.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 254 Aug 3 08:29 copy.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 85 Jul 27 08:29 f1.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 155 Jul 30 05:11 fork.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 374 Aug 4 05:45 fork3.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 366 Aug 4 06:08 fork4.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 319 Aug 27 04:57 infinateloop.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 240 Sep 15 05:58 malloc.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 254 Sep 15 06:00 malloc1.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 1 Aug 27 05:13 multiprocess.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 302 Aug 27 05:38 namedpipeex.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 416 Aug 27 05:49 namedpipeex2.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 315 Jul 30 06:00 open2.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 254 Aug 4 05:32 pipe1.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 19 Aug 8 09:29 pipe6.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 304 Aug 8 09:45 pipe7.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 578 Sep 15 07:04 posix.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 419 Aug 22 05:45 practical1.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 418 Aug 25 15:38 practical2.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 333 Sep 1 05:32 process1.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 274 Sep 1 05:16 process2.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 1043 Sep 16 15:55 producerconsumer.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 685 Sep 16 16:00 program5.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 87 Aug 3 06:48 read1.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 276 Aug 3 08:20 readwrite.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 1332 Aug 3 08:00 readwrite.c.save
-rw-r--r-- 1 haadi_hasil haadi_hasil 45 Aug 25 15:46 sample_practical
-rw-r--r-- 1 haadi_hasil haadi_hasil 199 Aug 31 08:02 signal.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 686 Aug 31 08:15 signal2.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 339 Aug 31 08:21 signal3.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 800 Aug 25 16:06 skill2.1.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 253 Aug 25 15:57 skill2.c
-rw-r--r-- 1 haadi_hasil haadi_hasil 11475 Aug 25 15:50 trace.txt

```